

Course- B. Tech
Course Code- CSET343
Year- 4th Year

Type- Specialization Elective
Course Name- AI in Healthcare
Semester- VII

Lab Assignment -6

Objective: The objective of this assignment is to design and evaluate CNN models for the automated diagnosis of heart disease.

Dataset: [Chest X-Ray Images](#)

- **Number of Instances:** ~5,863 images
- **Classes:** Normal, Pneumonia

Task 1: Data Acquisition and Exploration

- Download and load the dataset
- Display 5–10 random sample images from each class.
- Report number of images per class (class balance).

Task 2: Data Preprocessing

- Resize images to a fixed dimension (e.g., 128×128).
- Convert to grayscale if required.
- Apply feature extraction methods:
 - Histogram of Oriented Gradients (HOG)
 - Local Binary Patterns (LBP)
 - Gray-Level Co-occurrence Matrix (GLCM)
- Prepare feature vectors for each image.
- Split data into **training (70%)**, **validation (15%)**, and **testing (15%)**.

Task 3: Model Training and Basic Evaluation

- Implement **baseline models**: CNN Model.
- Tune hyperparameters using cross-validation.
- Evaluate models using **Accuracy, Precision, Recall, F1-score, and AUROC**.
- Create a **confusion matrix** and explain misclassifications.
- Compare classifiers and identify the best performer.